ModelIt!

Student Activity Pack

Can We Feed 10 Billion People?

Modeling Agricultural Systems, Land Use, and Global Food Security

Grade 12L2 | NGSS HS-LS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Where does your food come from? Can you trace a typical meal from farm to your plate and estimate how far it traveled?**

2. **What percentage of food produced globally do you think is wasted or lost before anyone eats it? Why does this happen?**

3. **What do you think is the biggest challenge in feeding a growing world population — producing more food, distributing it fairly, or both?**

4. **How do you think farming affects the environment? What trade-offs exist between food production and environmental health?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Global Population | Agricultural Land Area | Crop Yield Per Hectare | Soil Health Index | Water Availability for Agriculture | Food Waste and Loss | Biodiversity in Agricultural Landscapes

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Industrial Expansion

Agricultural Land: +20% | Chemical Inputs: Maximum | Soil Practices: Conventional Tillage

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Population Pressure

Global Population: 10 Billion | Agricultural Land: Current | Farming: Current Practices

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Regenerative Transition

Agricultural Land: Current | Waste Reduction: 50% | Soil Health: Improving | Biodiversity: Protected

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Dietary Composition — The proportion of animal versus plant-based foods in the average diet — producing one calorie of beef requires 25 calories of feed grain and 15,000 liters of water, so dietary shifts toward more plant-based foods could dramatically reduce the land and water footprint of the food system

Climate Change Impact on Agriculture — The effect of rising temperatures, changing precipitation, and extreme weather on crop yields — climate models project 10-25% yield declines in tropical and subtropical regions by 2050, precisely where population growth is highest

Agricultural Technology and Innovation — Advances in precision agriculture, vertical farming, gene editing, and satellite monitoring that can increase yields per hectare — technology can improve efficiency but cannot overcome the fundamental constraints of soil health, water availability, and ecosystem degradation

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Earth currently struggles to feed 8 billion people equitably, with 800 million going hungry while one-third of all food is wasted — so when the population hits 10 billion by 2050, can we produce enough food without destroying the ecosystems we depend on? Use evidence from your simulation to support your answers.

1. **How does your model illustrate the tension between maximizing short-term food production and maintaining long-term agricultural sustainability?**

2. **Why is food waste reduction potentially more impactful than increasing production? What does your model show about the scale of waste?**

3. **Water emerged as a critical constraint in your model. What are the implications for agricultural regions that depend on depleting groundwater?**

4. **How would shifting global diets toward more plant-based foods change the carrying capacity calculation in your model?**

5. **What does your model suggest about the claim that technology alone can solve the food crisis without changes to farming practices, distribution, or consumption?**

8 STEM Challenge: Design a Sustainable Food System for a Growing City

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YOUR ENGINEERING MISSION

Design a comprehensive food system plan for a rapidly growing city of 2 million people that maximizes food security while minimizing environmental impact, incorporating local agriculture, waste reduction, and equitable distribution.

THE SCENARIO

A rapidly growing city in a semi-arid region is projected to double its population from 1 million to 2 million within 20 years. Currently, 90% of its food is trucked in from distant industrial farms, 35% of food is wasted, local farmland is being paved for development, and the regional aquifer is declining by 2 meters per year. The city government has commissioned your team to design a food system that can feed 2 million people sustainably for at least 50 years. Your model data should guide decisions about local food production, water management, waste reduction, and distribution equity.

Engineering Design Thinking

1. HOW DOES YOUR FOOD SYSTEM PLAN ADDRESS WATER SCARCITY — THE MOST CRITICAL CONSTRAINT IDENTIFIED BY YOUR MODEL?

2. WHAT SPECIFIC WASTE REDUCTION STRATEGIES DOES YOUR PLAN IMPLEMENT, AND WHAT PERCENTAGE OF THE FOOD GAP DO THEY CLOSE?

3. HOW DOES YOUR PLAN ENSURE EQUITABLE FOOD ACCESS ACROSS ALL NEIGHBORHOODS, INCLUDING LOW-INCOME COMMUNITIES?

4. WHAT ROLE DOES SOIL HEALTH REGENERATION PLAY IN YOUR PLAN\'S LONG-TERM SUSTAINABILITY?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

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Explain your design using scientific evidence from the simulation:

Real Career Connection

Agricultural scientists and food systems researchers study how to produce more food with fewer resources while maintaining ecosystem health. They work for the USDA, universities, international organizations like the FAO, and agricultural technology companies, earning $60,000-$125,000/year. Urban agriculture designers who plan rooftop farms, vertical farms, and community gardens earn $55,000-$100,000/year. Food policy analysts who shape government nutrition programs and agricultural subsidies earn $65,000-$120,000/year.